

International Conference on Machine Learning and Data Engineering

Forensic Sketch to Real Image Using DCGAN

Sreedev Devakumar, Greeshma Sarath

Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Amritapuri, India

Abstract

Converting pencil sketches to real image is required in forensic analysis for identifying a person. In a reported crime, the testimony explains the physical features of the accused by recalling them from memory. As the human face is so diverse and variable, it's tough to predict the person from a forensic sketch expert's low-quality sketch. In this paper we propose a deeplearning model using DCGAN to convert the sketch to real image. The prepared sketch is used as input for training, and after a set number of epochs, our deep learning model learns and produces a realistic high-resolution image of the suspect. The face recognition system is a powerful tool which can be used for a wide range of purposes, from security to entertainment. Our proposed method's final average SSIM is 0.587 where as in the existing methods the average SSIM is 0.554.

© 2023 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: GAN; DCGAN; SSIM; CycleGAN; CGAN; patch GAN; Dense layer

1. Introduction

Due of the sophistication of visual perception in a vibrant and highly varied physical environment, traditional image processing approaches such as computer vision (CV) are still unable to process digital images efficiently. However, as machine learning ideas advance, traditional sketch processing [4] practices will be further improved by combining machine learning approaches with traditional sketch processing [7] techniques to make image processing easier and automated by having removed human interference.

AI-based facial recognition is a tearing technology that can keep up with the digital world's rapid technological advancements. It assists in the detection of criminals by allowing a neural network to recognize a person's face and provide pencil sketch pictures in response. Generation and matching of real photographs from a sketch copy can be done with this procedure. Matching of the generated image is compared with an existing database. This reveals

* Corresponding author. Tel.: 9495454277

E-mail address: sreedevd00@gmail.com



Fig. 1. Overall working of GAN

the identity proofs such as PAN, electoral Id, Bank passbook details, passport, etc[19]. Here, we are following the Generative Adversarial Network (GAN) technique of deep learning concept for this purpose[3] for generating synthetic images, we convert the sketch to the real image.

Overall working of GAN is shown in Fig 1. The generator is generating the real image in the initial epochs using the noise. Noise is the pixel code of the sketch given as input. During training generator generate images using the noise and is given to the discriminator. The discriminator [10] take two inputs(generator output and the real image corresponding to the input sketch) and determines the generated image is fake or real based on the original input dataset.

In terms of similarity index, the proposed methodology beats established techniques for photo synthesis (SSIM) [13]. Significantly, the synthetic Photos demonstrating our method have proven to be more useful in enhancing the accuracy of forensic sketch to real image matching. Sketches with more specific descriptions of face characteristics are produced by further techniques based on generative adversarial networks (GANs). These techniques primarily concentrate on photo-sketch synthesis, although they are suitable for reverse creation.

The existing techniques for sketch to image conversion uses machine learning models like convolutional neural network (CNN)[15][17] and GAN for converting sketches to real images. The quality of the generated images can be further be improved by innovative image processing algorithms. Significant geometric deformations and textural alterations between a face shot and its corresponding sketch are possible due to the diverse characteristics of photographs and drawings. For these reasons, sketch-based face recognition is a tough heterogeneous problem[14]. In this paper we use DCGAN [8] with a generator and two discriminators. First discriminator acts like a patch GAN and calculates the loss, D_G_loss of generated image with respect to the original. The Second discriminator will get activated after 30 epochs. It calculates the loss D_loss and it has an additional dense layer to help the generator [11] generating more accurate real images. The average SSIM of the proposed model is 0.587 where as the existing methods got a highest average SSIM of 0.554. Normal DCGAN has an identification rate of 0.95 whereas the identification rate of proposed DCGAN with added dense layer is 0.98.

The paper contains the following session: Section II summarizes the related works and representative methods of sketch to photo conversion and GANs. Section III provide more details of proposed methods and architecture details. In section IV describes the experimental results and analysis and Finally, conclusion is given in section V.

2. RELATED WORK

2.1. Face Photo-Sketch conversion

Shikang Yu provided a unique conditional CycleGAN[1]. We taking use of the possibilities in particular. CycleGAN and conditional GANs, and create a feature-level architecture loss to assure that the resultant face images are

of excellent quality sketches. Face recognition against a gallery of mugshot images is done and use the produced face photos as an alternative for face sketching. The adversarial loss's design is an essential aspect of the purpose since it influences whether the synthesizing photo (or drawing) can be as close to the original as conceivable. The suggested technique beats state-of-the-art approaches in sketch-to-photo translation when compared to SSIM, so according to experimental tests on the public-domain database CUFSS. They've observed that during network training, performance could vary tremendously over a few epochs. Jian Zhao is proposed for generating precise pictures, presented a method for creating a true image of a human face using a detailed drawing and textual description of the image, as well as facial attributes. To separate the sketch from the cropped real picture, the XDoG method is utilized. Face illusion super resolution enhancer seems to have an issue with sketch to face creation. The system is a combination [6] of an extraction of features network, a down sampling network, and an up-sampling network, all of which utilise guided filter to decrease network layers. Celebrity Face Attributes would be one of the datasets analyzed (CelebA). The activation function and tanh for facial reconstructing are Parameterized Exponential Linear Units (PReLU). The method has a precision of 16.306 in terms of PSNR and SSIM. For face sketch-to-photo conversion.

Recent research on face sketch synthesizing has just shown advanced findings. To synthesis sketching in the wild, Peng et al. developed a strong patch representation-based probabilistic graphical model [2]. Lu et al. expanded the prior network and concentrated on the preprocessing of input photographs in order to apply exemplar-based approaches to get better results. They were inspired by FCN-[20]based networks. To guarantee getting the identity-specific information and the preservation of common face structure in the synthesis, Zhang et al. offered their approaches [4], which were based on Markov random fields and the neural probabilistic graphical model, respectively.

To lead sketch-photo translation while keeping identification information, Ye Lin suggests a deep model based on GANs [5] with slight modification. To make a complete output, we augment U-Net by putting an extra skip connection [15] to the input picture as non-embedded features and extending a deconvolution layer and a down-sample layer to the end of the original network. This method is thought to supply additional identification data to the creation process. Aside from the standard discriminator employed in conditioned GANs. The author also brings a new one that is only focusing on the target domain. Both discriminators required paired photos given input. One demands a pair from two domains, while someone wants a pair from the same domain that contains a genuine picture and its replication. The generator [11] may learn more target domain styles since the addition discriminator's input always has a realistic sample.

2.2. Deep Convolutional Generative Adversarial Network

Deep Convolutional GAN (DCGAN) [9] was developed by Yen-Cheng Liu, which integrates a variational classifier (VAE) with a domain adversarial network. Across domains, the Yen-Cheng DCGAN may be trained semi-supervised. so that the acquired model may be used to handle with input drawings of different patterns. For image synthesis learning, DCGAN uses image couples from both the sketch and picture area. In order to improve the trained model's skills to withstand input drawings, During the training stage, we basically can use more sketch pictures (with the style of interest) in different styles. Our DCGAN used the architecture of a variational autoencoder (VAE) to accomplish picture inversion in addition to these different styles in a GAN based network [6]. We evaluated the efficacy and resilience of DCGAN using both qualitative and quantitative methods in the tests.

3. Proposed Method

In this work, we proposed DCGAN (Deep Convolutional Generative Adversarial Network) [8] for mapping the data from the sketch domain to the photo domain. This model includes one generator network (G) and two discriminator networks (D₁, D₂). The generator uses convolutions of the input sketch and generate photos. The first discriminator is a patch GAN which calculates the L1 loss or discriminator loss. Then generator [11] proceeds with this discriminator loss for the generation in the next epochs. The second discriminator will get activated after 30 epochs. Unlike the first case the second discriminator is a normal discriminator with an added dense layer. Here the second discriminator maps the target generated image and target photo. The normal DCGAN has shown less accuracy but in our case the architecture is modified by adding dense layers in the discriminator part. Thus our final model became more accurate for this mapping.

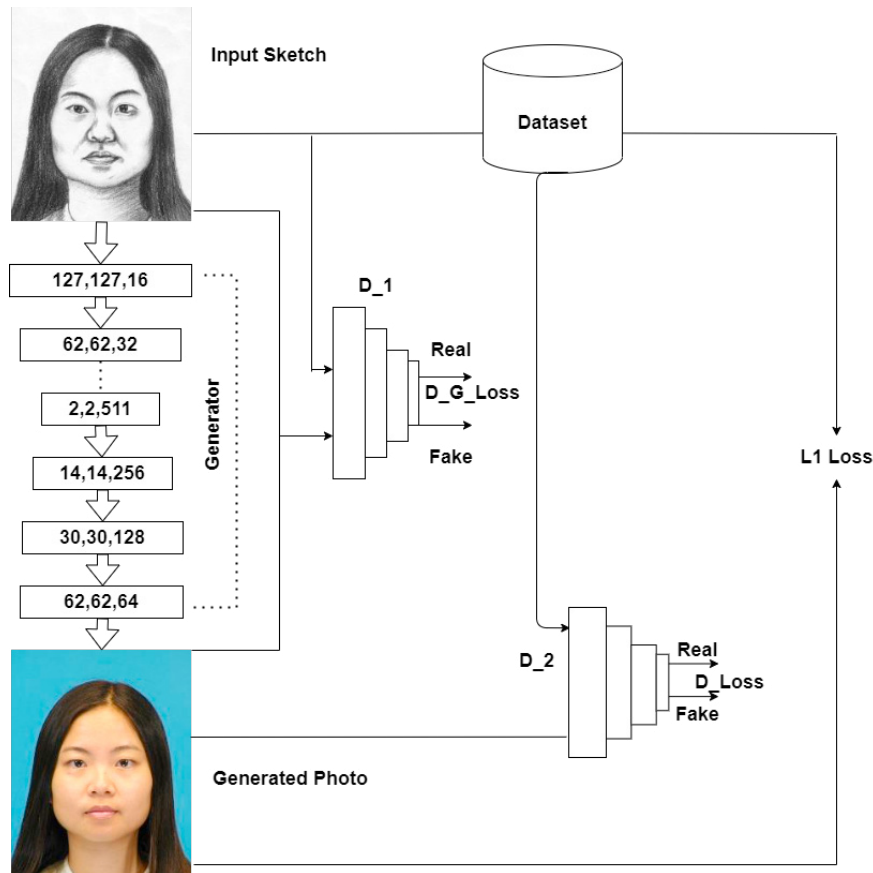


Fig. 2. Proposed DCGAN Architecture

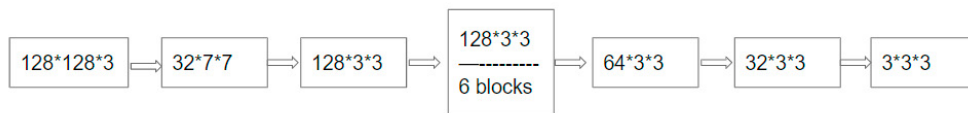


Fig. 3. Layer size structure of Generator

3.1. Network Architecture

This work contains one generator and two discriminators as shown in Fig 2. Both discriminators follow different architectures. One of them is a patch GAN, whereas the other one is a normal discriminator with an added dense layer. In generator conv2D, batch normalization and Relu layer is used. In discriminator conv2D, batch normalization and Leaky Relu layer is applied. In leaky Relu layer dropout function is set to 0.001. For each convolutions in discriminator 4x4 filters and stride=2 is used. At the same time the generator is convoluted with 3x3 filters and stride=2.

G Generator follows extended U-Net Architecture, where input is passed through skip connections and the obtained output is a 64 feature map. Here we used DCGAN with 6 batches of convolutions for generating better images. Initially our input is of size 128x128x3. After passing through first convolutional layer the size becomes 32x7x7. Three convolution layers are applied here shown in fig.3. After the third convolution the size of the output becomes 128x3x3. Relu activation function is used in this generator. We defined up to 6 blocks and each block consisting of

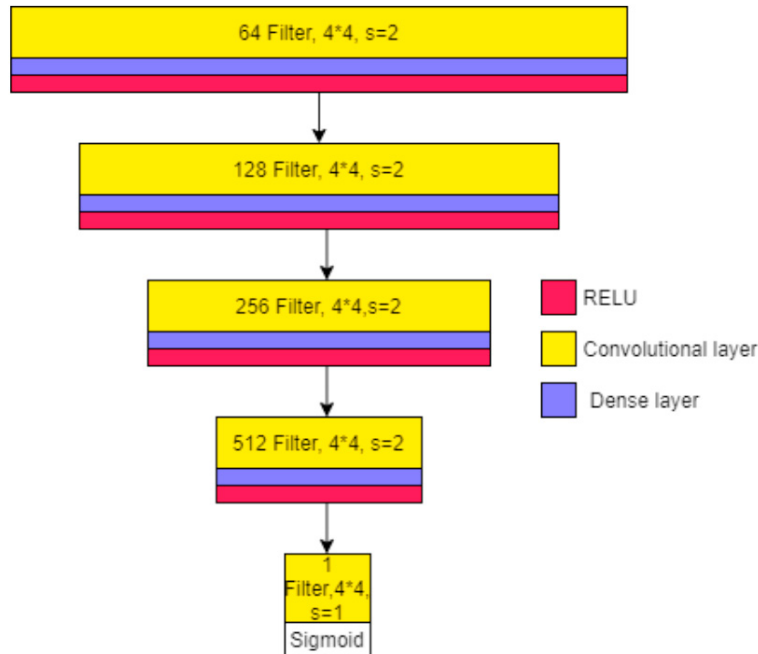


Fig. 4. Discriminator Architecture with adding Dense layer

conv2D(128x3x3), batch normalization and Relu activation function. Again this is added to two layers as v1 and v2. Likewise each layer has 2 blocks. Also used two conv2d transform. First conv2D [15] transform is 64,3x3 and other conv2D transpose of size 32,3x3 with same stride size 2. Through these convolutions we obtained the input sketch into generated output image.

Our architecture consist of 2 discriminators. First discriminator works based on patch Gan which is D_1. This D_1 is calculating the loss in initial stage(first 30 epochs) based on the generated output and corresponding original dataset. D_1 takes the generator output and executes patchGAN[12]. PatchGAN takes the input image and is transferred to a 128x128x3 dimensional array rather than a single scalar vector, which is essentially a convolutional network. The second discriminator(D_2) is a normal discriminator with an added dense layer. D_2 calculates the loss (L1 loss,d_loss) for generating better image from generator and down sampling the corresponding generated image for determining the los.

The proposed dense layer is added to the second discriminator (D_2). It consist of 4 layers of size 64,128,256,512 with 4x4 filter and stride=2. A Dense layer is added to this convolutions. Each dense layer has 2 layers and softmax activation function. This discriminator shows whether the image generated by the generator is fake or real. Discriminator or generator loss is produced here. The second discriminator(D_2) includes 4 convolution batch normalization layers. It follows the size as 64,128,256,512. Dense loss from this discriminator network layers helps the generator to understand more features of the sketch. v

4. Experimental Results

4.1. Dataset

The dataset used in this problem are CUHK Face Sketch FERET Database (CUFSF) [1] is a popular dataset for research on face photo-sketch synthesis and recognition. An artist created a drawing for each couple based on the color face shot that was taken in a frontal aspect under typical lighting conditions, and with a neutral expression. It offers 1,194 face pictures as well as 1,194 artist-drawn face drawings. We employ manually marked keypoints if facial

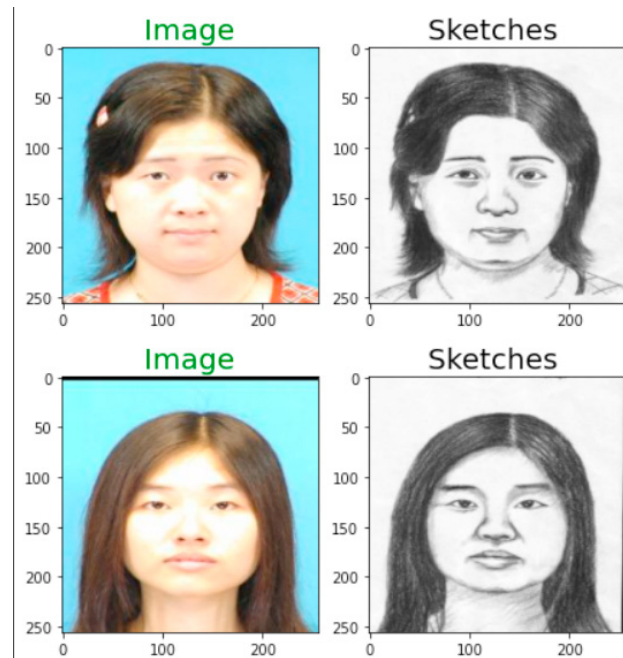


Fig. 5. Example for the Dataset Sketches Corresponding Images

Table 1. Average SSIM for Different Approaches

APPROACHES	SSIM
Original Sketch	0.2836
CycleGAN (UnPaired)	0.4796
CycleGAN (Paired)	0.5174
Pix2pix	0.5476
GAN With Feature Loss	0.5545
Proposed (DCGAN)	0.587

keypoints are not detectable. For network training, the cropped face pictures are resized to 64 64. When examining this shot, an artist has created a drawing with shape exaggeration for each person.

The testing set consists of 104 real image sketch pairs, whereas the training set consists of the remaining 1400 real image sketch pairings. We use the methodology to simulate drawings, which uses Sobel edge detection and contrast enhancement to construct the sketch images of the preceding face images, and the resulting (simulated) sketch and photo pairs are used as training data. The CUFSF sketches, on the other hand, were designed to include many distortions and exaggerations in order to resemble real-world sketches.

Based on the output obtained from two models of DCGAN, one with a dense layer and the other without a dense layer as shown in Fig.6. Experimental results show that output from the DCGAN with a dense layer clearly depicts the eye, nose, cheeks, facial structure, and hair color. Compared to the other model output of DCGAN with dense layers results in a quality image. SSIM evaluation method was used here which shows comparatively good value than other variants of GAN as shown in the Fig.6 is obtained. For further research studies can also conducted to increase the accuracy in the architecture of our model by adjusting L1 loss, discriminator loss.



Fig. 6. Generated Output without Adding Dense Layer

4.2. Stimulation Details

The learning rate used in this architecture is 0.001. We use Google colab pro platform for training and validation purpose and VS Code for local development. Output images are stored locally in validation time for calculating SSIM. In this problem the dataset size is small so traditional performance metrics may produce unreliable results. since we are using different algorithms, it's performance can be measured using the metric rank. Fig 8 displays the accuracy for face identification from Rank-1 to Rank-10 using various verifies. Our approach outperforms the competition in terms of identification information preservation.

4.3. Evaluation Metrics

We are used Structural Similarity Index (SSIM) is a perceptual metric to determine image resolution impairment. The SSIM index is a measure for analyzing how similar two pictures [14] are. The SSIM values range from 0 to 1, with 1 signifying a perfect match between the reconstituted picture and the original. Table I shows the average SSIM of the various techniques.

The cumulative match characteristic (CMC) is a way of displaying how accurately a biometric system performs when doing a closed-set identification job. We test several real image generation techniques from sketch. Generated real images are used to determine whether they can increase face identification accuracy. The lowest matching rate is provided by directly comparing the probe drawings to the gallery pictures, as predicted. The image generated by the proposed method is giving higher accuracy compared to images generated using Pix2Pix[18], CycleGAN [2]. The CMC curve for the different techniques is shown in fig 8.

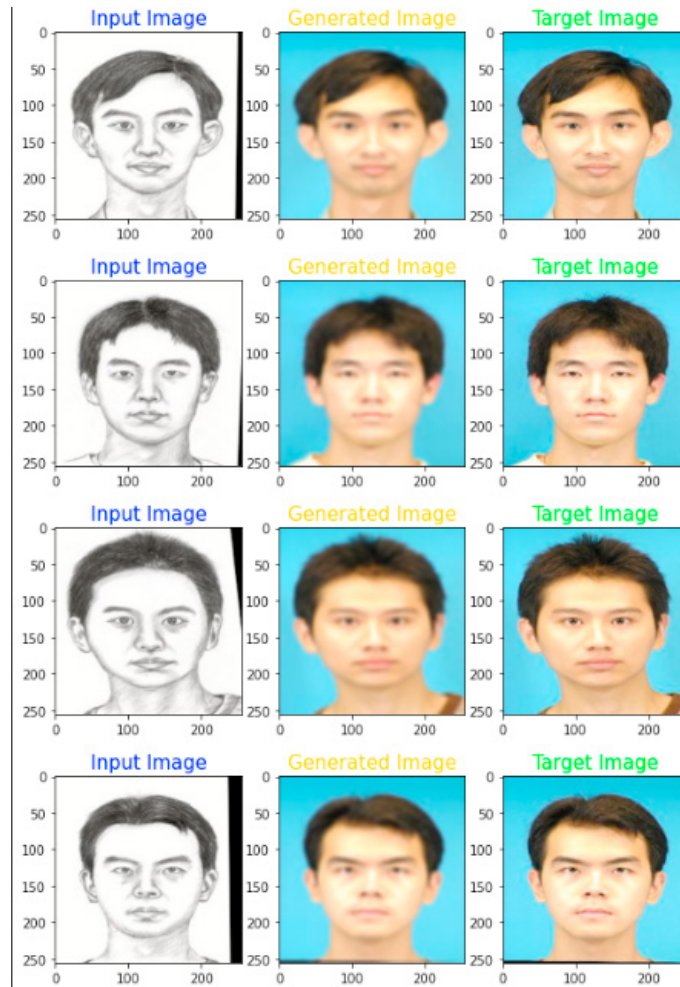


Fig. 7. Generated Output with Adding Dense Layer(Proposed Approach)

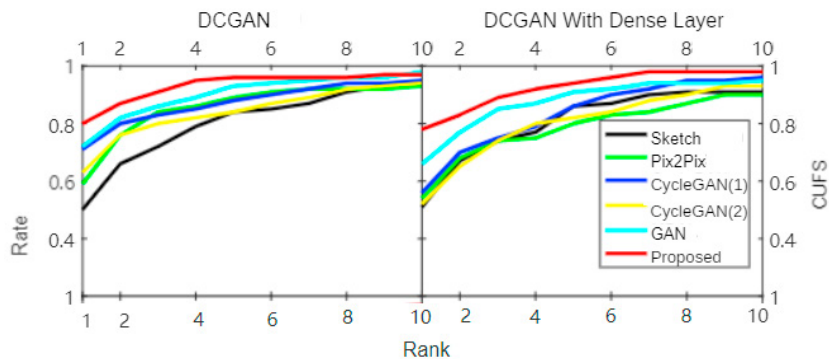


Fig. 8. CMC curve of face identification using generated images by the proposed method and comparing other methods

CONCLUSION

Several [2] existing techniques for converting sketches to real images are studied. We proposed, implemented and evaluated a novel technique for converting pencil sketches to high quality real images using DCGAN with one generator and two discriminators. First discriminator is a patch GAN whereas the second discriminator has an added dense layer to improve the accuracy. While comparing our experimental results with the existing approaches it's been shown that our proposed approach outperforms the traditional methods such as GAN, CycleGAN, CGAN in terms of accuracy.

The proposed method has been compared with the existing methods in terms of SSIM and identification rate. After the addition of dense layer, we noticed that the SSIM and identification rate of the proposed method is higher in comparison with the previous techniques. Our model performs efficiently with the dataset we have used. Our proposed architecture is entirely different from the previous traditional ones and also, we have changed various parameters too to achieve higher accuracy so it's been proved that proposed model is giving us much higher accuracy.

References

- [1] Yu, Shikang, et al. "Improving face sketch recognition via adversarial sketch-photo transformation." 2019 14th IEEE International Conference on Automatic Face Gesture Recognition (FG 2019). IEEE, 2019.
- [2] Yu, Shikang, et al. "Improving face sketch recognition via adversarial sketch-photo transformation." 2019 14th IEEE International Conference on Automatic Face Gesture Recognition (FG 2019). IEEE, 2019.
- [3] Liu, Yen-Cheng, et al. "Domain-Adaptive generative adversarial networks for sketch-to-photo inversion." 2017 IEEE 27th International Workshop on Machine Learning for Signal Processing (MLSP). IEEE, 2017.
- [4] Galea, Christian, and Reuben A. Farrugia. "Forensic face photo-sketch recognition using a deep learning-based architecture." IEEE Signal Processing Letters 24.11 (2017): 1586-1590.
- [5] Lin, Ye, et al. "An identity-preserved model for face sketch-photo synthesis." IEEE Signal Processing Letters 27 (2020): 1095-1099.
- [6] Yeshaswy, R., et al. "Design and analysis of a 6 Watt GaN based X-band power amplifier." 2016 International Conference on Communication and Signal Processing (ICCSP). IEEE, 2016.
- [7] Sarathkrishna, S., et al. "Design and analysis of GaN HEMT based LNA with CPW matching." 2014 Eleventh International Conference on Wireless and Optical Communications Networks (WOCN). IEEE, 2014.
- [8] Bushra, S. Nikkath, and K. Uma Maheswari. "Crime Investigation using DCGAN by Forensic Sketch-to-Face Transformation (STF)-A Review." 2021 5th International Conference on Computing Methodologies and Communication (ICCMC). IEEE, 2021.
- [9] Li, Qiyue, et al. "AF-DCGAN: Amplitude feature deep convolutional GAN for fingerprint construction in indoor localization systems." IEEE Transactions on Emerging Topics in Computational Intelligence 5.3 (2019): 468-480.
- [10] Schonfeld, Edgar, Bernt Schiele, and Anna Khoreva. "A u-net based discriminator for generative adversarial networks." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020.
- [11] Gnanha, Aurele Tohokantche, et al. "The residual generator: An improved divergence minimization framework for GAN." Pattern Recognition 121 (2022): 108222.
- [12] Demir, Ugur, and Gozde Unal. "Patch-based image inpainting with generative adversarial networks." arXiv preprint arXiv:1803.07422 (2018).
- [13] Channappayya, Sumohana S., Alan Conrad Bovik, and Robert W. Heath. "Rate bounds on SSIM index of quantized images." IEEE Transactions on Image Processing 17.9 (2008): 1624-1639.
- [14] Klare, Brendan, Zhifeng Li, and Anil K. Jain. "Matching forensic sketches to mug shot photos." IEEE transactions on pattern analysis and machine intelligence 33.3 (2010): 639-646.
- [15] MS, Vishnu Mohan, and Vivek Menon. "Measuring Viscosity of Fluids: A Deep Learning Approach Using a CNN-RNN Architecture." The First International Conference on AI-ML-Systems. 2021.
- [16] Mohan, Vishnu. "Detection of COVID-19 from Chest X-ray Images: A Deep Learning Approach." 2021 Ethics and Explainability for Responsible Data Science (EE-RDS). IEEE, 2021.
- [17] N. Aloysius and M. Geetha, "A review on deep convolutional neural networks", in 2017 International Conference on Communication and Signal Processing (ICCSP), Chennai, India, 2017.
- [18] V. Balachandran and S. Sarath, "A Novel Approach to Detect Unmanned Aerial Vehicle using Pix2Pix Generative Adversarial Network," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), 2022, pp. 1368-1373, doi: 10.1109/ICAIS53314.2022.9742902.
- [19] Greeshma Sarath, R Jayapriya "Securing remote database server using fully homomorphic encryption and re-encryption" Security in Computing and Communications: Third International Symposium 2015.
- [20] Archana, M. C. P., and J. Vinitha Panicker. "Deep Convolutional Neural Networks for Multiclass Cervical Cell Classification." 2022 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET). IEEE, 2022.